

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}}$$

$$\bar{a}'_{\text{true}}(\bar{w}) > 0 \quad \text{on} \quad \bar{w} > 1 \quad \implies \quad \bar{a}' = \frac{(1 - \bar{a})^2}{b_{\text{true}}(\bar{a})} \quad \text{exactly,} \quad b_{\text{true}} := \frac{(1 - \bar{a}_{\text{true}})^2}{\bar{a}'_{\text{true}}}$$

$$\begin{aligned} \bar{a}_{\text{true}}(1) = 0, \quad \bar{a}'_{\text{true}}(1) = 1 &\implies b_{\text{true}}(\bar{a} = 0) = 1 \\ 1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}) &\implies b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{9}{8} \end{aligned}$$

$$\begin{aligned} \bar{a}'(\bar{a}) &:= \frac{d\bar{a}}{d\bar{w}} = (1 - \bar{a})^2 \quad (b_{\text{true}}(\bar{a}) \rightarrow 1, \text{ its contact value}) \\ a_w &= a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R} \end{aligned}$$

$$\frac{\bar{a}'(\bar{a}_{\text{true}})}{\bar{a}'_{\text{true}}} = b_{\text{true}} = \frac{9}{8} + \frac{1}{2\bar{w}^2} - \frac{57}{50\bar{w}^3} + \mathcal{O}(\bar{w}^{-4})$$

2 Jacobians

$$\frac{\partial \bar{a}'}{\partial \bar{a}} = -2(1 - \bar{a}), \quad \frac{\partial}{\partial \delta a} \left[\bar{a}_w + (1 - \bar{a}_w)^2(\cdot) + \delta a \right] = 1$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\ \bar{w}_d[k + 1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\ \delta \bar{w}_1[k + 1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k + 1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k + 1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \end{aligned}$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i]$$

$$\varepsilon_w[k] = \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] + (1 - \lambda_c) n_w[k]$$

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

(a)

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

(b)

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] + \delta a[k]$$

$$\delta a[k] = \left[\bar{a}'_{\text{true}}[k] - (1 - \bar{a}_w[k])^2 \right] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

$$\delta a[k + 1] = \delta a[k]$$

4 Process model

$$(a) \quad \mathbf{x} = \left[\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w \right]^\top$$

$$(b) \quad \mathbf{x} = \left[\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, \delta a \right]^\top \quad (d = 2)$$

5 Estimator

$$\delta \hat{w}_1[k + 1] = \delta \hat{w}_2[k]$$

$$\delta \hat{w}_2[k + 1] = \delta \hat{w}_3[k]$$

$$\delta \hat{w}_3[k + 1] = \lambda_c \delta \hat{w}_3[k]$$

$$\begin{aligned}
\text{(a)} \quad \hat{a}_w[k+1] &= \hat{a}_w[k] + (1 - \hat{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right] \\
\text{(b)} \quad \hat{a}_w[k+1] &= \hat{a}_w[k] + (1 - \hat{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right] + \delta \hat{a}[k] \\
\text{(b)} \quad \delta \hat{a}[k+1] &= \delta \hat{a}[k]
\end{aligned}$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned}
e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k] \\
e_{\delta \bar{w}_2}[k+1] &= e_{\delta \bar{w}_3}[k] \\
e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \left[1 + (1 - \hat{a}_w)^2 F_{dw} - 2(1 - \hat{a}_w)(\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) \right] e_{\bar{a}_w}[k] \\
&\quad + (1 - \lambda_c)(1 - \hat{a}_w)^2 e_{\delta \bar{w}_3}[k] + (1 - \hat{a}_w)^2 \varepsilon_w[k] \quad [+ e_{\delta a}[k] \text{ in (b)}] \\
e_{\delta a}[k+1] &= e_{\delta a}[k]
\end{aligned}$$

$$\text{(b)} \quad F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c)(1 - \hat{a}_w)^2 & 1 + (1 - \hat{a}_w)^2 F_{dw} - 2(1 - \hat{a}_w)(\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(a) is the leading 4×4 block.

$$\mathbf{q}[k] = [0, 0, -\varepsilon_w, (1 - \hat{a}_w)^2 \varepsilon_w, 0]^\top$$

$$e_{\bar{a}}(\bar{w}) = -\bar{a}'(\bar{w}) e_{\bar{w}_0} \propto \frac{1}{(\bar{w} - \bar{w}_0)^2}$$

7 Measurement and innovation

$$\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k]
\end{aligned}$$

$$\bar{a}_w[k-d] = \bar{a}_w[k] - (1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k] \quad [-d \delta a[k] \text{ in (b)}]$$

$$\begin{aligned}\frac{\partial y_2}{\partial \bar{a}_w} &= 1 + 2(1 - \hat{\bar{a}}_w) \nabla_d \bar{w}_d \\ \frac{\partial y_2}{\partial \delta a} &= -d\end{aligned}$$

$$(b) \quad H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2(1 - \hat{\bar{a}}_w) \nabla_d \bar{w}_d & -d \end{bmatrix}$$

$$K[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$\begin{aligned}e_{y_1}[k] &= e_{\delta \bar{w}_1}[k] + n_w[k] \\ e_{y_2}[k] &= \left[1 + 2(1 - \hat{\bar{a}}_w) \nabla_d \bar{w}_d \right] e_{\bar{a}_w}[k] - d e_{\delta a}[k] + n_a[k]\end{aligned}$$

8 Process and measurement noise

$$\begin{aligned}q_3 &= -\bar{a}_w \bar{f}_T \\ q_4 &= -(1 - \hat{\bar{a}}_w)^2 q_3 \\ \gamma(\bar{w}) &= \gamma_N c(\bar{w}) \\ \sigma_{\bar{f}_T}^2 &= a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t}\end{aligned}$$

$$\begin{aligned}Q_{33} &= \frac{4k_B T a_o}{R} \bar{a}_w \\ Q_{34} = Q_{43} &= -(1 - \hat{\bar{a}}_w)^2 Q_{33} \\ Q_{44} &= (1 - \hat{\bar{a}}_w)^4 Q_{33} \\ Q_{55} &= 0\end{aligned}$$

$$\begin{aligned}R &= \text{diag}(R_1, R_2) \\ R_1 &= \sigma_{n_w}^2 \\ R_2 &= K_{\text{var}} I F_{\text{var}} (\bar{a}_w + \xi)^2 + d Q_{44} \\ K_{\text{var}} &= \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}} \\ \sigma_{\delta \bar{w}_r}^2 &= C_{dpmr} \frac{4k_B T a_o}{R} (\bar{a}_w + \xi) \\ \xi &= \frac{C_n \sigma_{n_w}^2}{C_{dpmr} (4k_B T a_o / R)}\end{aligned}$$

$$Q_{55}^* = \text{Var}(\delta a[k+1] - \delta a[k])$$

Seed and prior

$$\delta \hat{a}[0] = 0$$

$$\hat{a}_w[0] = 1 - \frac{1}{\bar{w}[0] - \hat{w}_0}$$

$$\left. \frac{\partial \bar{a}}{\partial \delta a} \right|_{\bar{w}} = \frac{1 - \bar{a}}{1 + \delta a}$$

$$\left. \frac{\partial \bar{a}}{\partial \bar{w}_0} \right|_{\bar{w}} = -\bar{a}'$$

$$P_{44}[0] = \left(\hat{a}'[0] \right)^2 P_{\bar{w}_0} + \text{floor}_a^2$$

$$P_{45}[0] = 0$$

$$P_{55}[0] = P_{\delta a}$$